COMPOSITE INDICATORS ANALYSIS

Exploratory Cross-Country Analysis of Digital Connectivity and Infrastructure

1. Overview

The composite indicator provides a comparative measure of digital connectivity and enabling infrastructure across 180 countries. It combines four indicators—access to electricity, fixed broadband subscriptions, individuals using the Internet, and mobile cellular subscriptions. The analysis examines cross-country patterns and shows how different dimensions contribute to overall composite performance.

2. Methodology

The four indicators are converted to a common 0–1 scale using min–max normalization. Each indicator receives an equal weight of 0.25, and the weighted values are aggregated to obtain the composite score. Countries are ranked in descending order of the composite score. The approach facilitates comparison across indicators measured in different units.

Component	Indicator	Weight
I1	Access to electricity (% of population)	0.25
I2	Fixed broadband subscriptions (per 100 people)	0.25
I3	Individuals using the Internet (% of population)	0.25
I4	Mobile cellular subscriptions (per 100 people)	0.25

3. Overall Findings

The mean composite score is 0.519, while the median is 0.574. The spread of scores indicates substantial cross-country variation in digital connectivity and enabling infrastructure. The ranking also shows that strong performance in one dimension does not necessarily imply equally strong performance across all four dimensions.

Highest-ranked countries

Rank	Country	Composite Score	Interpretive observation
1	Hong Kong SAR, China	0.825	High relative performance across the selected dimensions.
2	Monaco	0.787	High relative performance across the selected dimensions.
3	Andorra	0.781	High relative performance across the selected dimensions.
4	Korea, Rep.	0.778	High relative performance across the selected dimensions.
5	Liechtenstein	0.769	High relative performance across the selected dimensions.

6	United Arab Emirates	0.766	High relative performance across the selected dimensions.
7	Saudi Arabia	0.761	High relative performance across the selected dimensions.
8	Switzerland	0.753	High relative performance across the selected dimensions.
9	Denmark	0.742	High relative performance across the selected dimensions.
10	China	0.740	High relative performance across the selected dimensions.

The top of the ranking includes Hong Kong SAR, China; Monaco; Andorra; Korea, Rep.; Liechtenstein; the United Arab Emirates; Saudi Arabia; Switzerland; Denmark; and China. These results describe performance on the selected indicators and are not a general ranking of national development.

4. India and Selected Countries

India is considered alongside selected Asian, emerging-market and regional comparator countries to contextualize its position within the wider dataset. The comparison demonstrates differences in composite performance across countries with varied connectivity profiles.

Country	Composite Score	Rank	Position relative to India
China	0.740	10	Higher composite score
Viet Nam	0.612	71	Higher composite score
Malaysia	0.609	73	Higher composite score
Brazil	0.602	79	Higher composite score
South Africa	0.508	114	Higher composite score
Indonesia	0.494	117	Higher composite score
Sri Lanka	0.465	124	Higher composite score
Bangladesh	0.447	126	Higher composite score
India	**0.445**	**128**	Reference country
Pakistan	0.403	139	Lower composite score

India records a composite score of 0.445 and ranks 128 out of 180 countries. Within the selected comparison group, India is positioned below China, Viet Nam, Malaysia, Brazil, South Africa, Indonesia, Sri Lanka and Bangladesh, while Pakistan has a lower composite score. The comparison is descriptive and reflects only the four indicators included in the composite.

5. Dimension-Level Analysis

The aggregate score becomes more informative when the component indicators are examined separately. The normalized profile highlights differences between basic infrastructure access, fixed connectivity and actual internet use.

Indicator	India: Normalized Score	Analytical reading
Access to electricity (% of population)	0.999	Very high relative position within the observed range.

Fixed broadband subscriptions (per 100 people)	0.057	Lowest normalized component; indicates a relative fixed-connectivity constraint.
Individuals using the Internet (% of population)	0.616	Moderate relative position, indicating substantial but incomplete internet adoption.
Mobile cellular subscriptions (per 100 people)	0.107	Lower normalized position; subscription figures require contextual interpretation because subscriptions can exceed unique users/population.

6. Key Patterns from the Exploratory Analysis

- Digital connectivity is multidimensional: the four indicators capture different aspects of the digital ecosystem.
- Countries can reach similar composite scores through different combinations of strengths, so the aggregate ranking can conceal distinct country profiles.
- Fixed broadband and internet use provide complementary information; infrastructure availability does not automatically translate into actual usage.
- Mobile cellular subscriptions measure subscriptions rather than unique individuals and can exceed 100 per 100 people, requiring careful interpretation.
- Indicator-level decomposition provides more policy-relevant information than the composite rank alone.

7. Policy Implications

The composite can serve as an entry point for policy diagnosis and further investigation. The results suggest four broad analytical areas:

Connectivity infrastructure: Assess fixed and last-mile connectivity, network coverage, reliability and quality.

Affordability and meaningful access: Examine affordability, device availability, digital skills and the ability to use digital services effectively.

Digital inclusion: Disaggregate national averages by rural–urban location, income, gender and age where data permit.

Digital outcomes: Explore links between connectivity and outcomes such as education, employment, financial inclusion and use of digital public services.

8. Limitations and Further Analysis

- Equal weighting is a methodological choice; sensitivity analysis with alternative weights can test the robustness of country rankings.
- Min–max normalization is influenced by the observed minimum and maximum values, particularly where extreme observations are present.
- The composite is a relative benchmarking measure and does not capture every dimension of digital development.
- The cross-sectional analysis does not establish trends or causal relationships.
- Future work can incorporate time-series analysis, peer-group benchmarking, distributional analysis, robustness checks and qualitative evidence.

9. Conclusion

The composite indicator reveals considerable cross-country variation in digital connectivity and enabling infrastructure. Its main analytical value lies in combining an overall benchmark with component-level interpretation. India's score of 0.445 and rank of 128 illustrate how aggregate performance can coexist with substantial variation across individual dimensions. The results therefore provide a useful baseline for comparative analysis and for identifying areas that merit deeper quantitative and qualitative investigation.

Source: World Development Indicators (WDI), 2024; calculations based on the selected indicators and composite methodology.

Link of excel file-
https://1drv.ms/x/c/f0cd9ed6c5377fba/IQAZLcmmNqo1SZ0B4Hr6JL2WAXVKKu4qTKnwZ8vlM8YrYKs